KTH Computer Science and Communication

Exam in DD2421 Machine Learning
2024-10-24, kl 8.00 – 12.00

Aids allowed: *calculator*, *language dictionary*.

To take this exam you must be registered to this specific exam as well as to the course.

In order to pass this exam, your score x first needs to be 20 or more (out of 42, full point). In addition, given your points y from the Programming Challenge (out of 18, full point), the requirements on the total points, $p = x + y$, are preliminarily set for different grades as:

$$
\begin{aligned}
54 < p \leq 60 &\quad \rightarrow \quad A \\
48 < p \leq 54 &\quad \rightarrow \quad B \\
42 < p \leq 48 &\quad \rightarrow \quad C \\
36 < p \leq 42 &\quad \rightarrow \quad D \\
29 < p \leq 36 &\quad \rightarrow \quad E \quad \text{(A pass is guaranteed with the required points for 'E'.)} \\
0 \leq p \leq 29 &\quad \rightarrow \quad F
\end{aligned}
$$

This exam consists of sections **A**, **B**, and **C**.
NB: use different papers (answer sheets) for different sections A, B, and C.
This means you shall start a new page after section A, and again after section B.

A Graded problems

Potential inquiries to be addressed to Atsuto Maki.

A-1 Terminology (4p)

For each term (a–d) in the left list, find the explanation from the right list which *best* describes how the term is used in machine learning.

a) k-means

b) RANSAC

c) k-fold cross validation

d) k-nearest neighbour

1) Robust method to fit a model to data with outliers

2) An ensemble approach by generating k different models

3) Method for estimating the mean of k observations

4) Clustering method based on centroids

5) An efficient method to find the optimal trade-off

6) Class prediction by a majority vote

7) Strategy to completely search a space while causing mess

8) A technique for assessing a model while exploiting available data for training and testing

Solution: a-4, b-1, c-8, d-6

A-2 Nearest Neighbor, Classification (4p)

Suppose that we take a data set, divide it into two parts of equal size, Part I and Part II. We try out two different classification procedures, by using Part I and Part II as our training set and test set, respectively, That is, we use half of the data for training, and the remaining half for testing.

a) First we use 1-Nearest Neighbor rule (1-NN) and get an average error rate (averaged over both test and training data sets) of 8%. What was the error rate with 1-nearest neighbor on the test set? Briefly reason the answer. (1p)

b) Next we use the Adaboost Algorithm and get an error rate of 10% on the training data. We also get the average error rate (averaged over both test and training data sets) of 12%. What was the error rate with the Adaboost Algorithm on the test set? (1p)

c) Now, we swap the roles of Part I and Part II, and repeat the same experiments. On the test set (Part I), we get an error rate of 12% with both 1-NN and the Adaboost Algorithm. Based on all these results, by the cross-validation, indicate the method which we should prefer to use for classification of new observations, with a simple reasoning. (2p)

Solution:

a) 16%. (Training error for 1-NN is always zero, and therefore the testing error is 16%.)

b) 14%.

c) Adaboost because it achieves lower error rate on the test data on average (13% < 14%).

A-3 Entropy and Decision Trees/Forests (5p)

a) Consider a single toss of *skewed* coin (it is likely to show one side more than the other side). Regarding the uncertainty of the outcome {head, tail}, which one of the following is correct?

i. the entropy is greater than one bit.
ii. the entropy is equal to one bit.
iii. the entropy is smaller than one bit.

Simply indicate your choice. (1p)

b) Indicate any valid ones as the basic strategy for selecting a question (attribute) at each node in decision trees.

i. To minimize the expected reduction of the entropy.
ii. To minimize the expected reduction of gini impurity.
iii. To maximize the expected reduction of the entropy.

Simply indicate your choice. (1p)

c) Draw a decision tree of *depth two* that is consistent with the training data in the table. $a_1, ..., a_4$ are attributes of the training data. (It is not required to use information gain to solve the task.) (2p)

a_1	a_2	a_3	a_4	$class$
0	0	0	0	-
0	0	1	1	+
0	1	0	1	+
0	1	1	0	-
1	0	0	0	+
1	0	1	1	+
1	1	1	0	-
1	1	0	1	-

d) Suppose we have trained a Decision Forest using five bootstrapped samples from a data set containing three classes, {Red, Yellow, Green}. We then applied the model to a specific test input, $\mathbf{x}$, and observed five estimates of class distributions from the five trees, respectively:

$\{P_1(\text{Red}|\mathbf{x}), P_1(\text{Yellow}|\mathbf{x}), P_1(\text{Green}|\mathbf{x})\}$ = {0.1, 0.5, 0.4}
$\{P_2(\text{Red}|\mathbf{x}), P_2(\text{Yellow}|\mathbf{x}), P_2(\text{Green}|\mathbf{x})\}$ = {0.2, 0.3, 0.5}
$\{P_3(\text{Red}|\mathbf{x}), P_3(\text{Yellow}|\mathbf{x}), P_3(\text{Green}|\mathbf{x})\}$ = {0.3, 0.6, 0.1}
$\{P_4(\text{Red}|\mathbf{x}), P_4(\text{Yellow}|\mathbf{x}), P_4(\text{Green}|\mathbf{x})\}$ = {0.2, 0.4, 0.4}
$\{P_5(\text{Red}|\mathbf{x}), P_5(\text{Yellow}|\mathbf{x}), P_5(\text{Green}|\mathbf{x})\}$ = {0.2, 0.3, 0.5}

Let us consider the average probability for combining these results into a single class prediction. What will the final class prediction be? *Motivate your answer by short phrases*. (1p)

Solution: **a**)-iii. **b**)-iii. **c**)-drawing. **d**)-Yellow, (0.2,0.42,0.38)

A-4 Regression with regularization (5p)

a) In regression, regularization is a process of introducing additional term, so-called shrinkage penalty. Which ones of the three methods includes the additional term.

i. Logistic regression.
ii. Ridge regression.
iii. k-NN regression.

Simiply indicate your choice. (1p)

b) For a set of N training samples $\{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\}$, each consisting of input vector $\mathbf{x}$ and output y, suppose we estimate the regression coefficients $\mathbf{w}^\top (\in \mathbf{R}^d) = \{w_1, \ldots, w_d\}$ in a linear regression model by minimizing

$$\sum_{n=1}^{N} (y_n - \mathbf{w}^\top \mathbf{x}_n)^2 \quad \text{subject to} \quad \sum_{i=1}^{d} |w_i| \leq s$$

for a particular value of s. As we *increase* s from 0, the trainig error and test errors (residual sum of squares, RSS) are believed to change. Indicate which of {i,ii,iii,iv,v} is correct about the *test* error, and *justify your answer*. (2p)

i. Remain constant.
ii. Steadily decrease.
iii. Steadily increase.
iv. Decrease initially, and then eventually start increasing in a U shape.
v. Increase initially, and then eventually start decreasing in an inverted U shape.

c) As we *increase* s from 0, what happens to the variance. Choose the answer from the five alternatives above, and *justify your answer*. (1p)

d) The method described in **b**) is called LASSO and known to yield *sparse models*. Briefly explain what property of it enables the sparcity in a short sentence. (1p)

Solution:

a) ii.

b) iv. When $s = 0$, all w_i are zero; the model is extremely simple, predicting a constant and has no variance with the prediction being far from actual value (thus with high bias). As we increase s, all w_i increase from zero toward their least square estimate values while steadily decreasing the *training* error to the ordinary least square RSS (and also decreasing bias as the model continues to better fit training data). The values of w_i then become more dependent on training data, thus increasing the variance. The *test* error initially decreases as well, but eventually start increasing due to overfitting to the training data.

c) iii.

d) The variable selection property.

A-5 Dimensionality Reduction, Subspace Methods (4p)

We consider to solve a K-class classification problem with the Subspace Method and for that we compute a subspace $\mathcal{L}^{(j)}$ $(j = 1, ..., K)$ using training data for each class, respectively. That is, given a set of feature vectors (as training data) which belong to a specific class C (i.e. with an identical class label), we perform PCA on them and generate an orthonormal basis $\{\mathbf{u}_1, ..., \mathbf{u}_p\}$ which spans a p-dimensional subsapce, $\mathcal{L}$, as the outcome.

Provide an answer to the following questions.

a) Given that we compute $\{\mathbf{u}_1, ..., \mathbf{u}_p\}$ as eigenvectors of the auto-correlation matrix Q based on the training data, how should we choose the eigenvectors in relation to the corresponding eigenvalues of Q? (1p)

b) We have a new input vector $\mathbf{x}$ whose class is unknown, and consider its projectiton length on each $\mathcal{L}^{(j)}$. Describe how the projection length is represented, using a simple formula. (2p)

c) Briefly explain the way to determine the class to which vector $\mathbf{x}$ should belong using the projection lengths. (1p)

Solution:

a) Choose the eigenvectors corresponding to the largest eigenvalues of Q.

b) $\sqrt{S}$ where $S = \Sigma_{i=1}^{p}(\mathbf{x}, \mathbf{u}_i)^2$

c) $\mathbf{x}$ should belong to the class where the projection length to the corresponding subspace is the maximum.

B Graded problems

Potential inquiries to be addressed to Olov Andersson.

B-1 Probability Theory (4p)

You are training a binary classifier to detect whether an email is spam or not. The following probabilities are given:

- $P(\text{Spam}) = 0.3$ – The probability that a randomly selected email is spam.
- $P(\text{Offer} \mid \text{Spam}) = 0.8$ – The conditional probability that an email contains the word "Offer" given that it is spam.
- $P(\text{Offer} \mid \neg\text{Spam}) = 0.2$ – The conditional probability that the email contains the word "Offer" given that it is NOT spam.
- $P(\text{Urgent} \mid \text{Spam}) = 0.6$ – The conditional probability that the mail contains the word "Urgent" given that it is spam.
- $P(\text{Urgent} \mid \neg\text{Spam}) = 0.1$ – The conditional probability that the mail contains the word "Urgent" given that it is NOT spam.

Using Bayes' Theorem, answer the following:

a) What is the probability that an email containing the word "Offer" is spam? (**Calculate** $P(\text{Spam} \mid \text{Offer})$) (2p)

b) Suppose the classifier also checks for the word "Urgent" in emails. Now **calculate** $P(\text{Spam} \mid \text{Offer}, \text{Urgent})$. You may assume that the random variables Offer and Urgent are independent. (2p)

Solution:

Solution:

a) Using **Bayes' Theorem**, the probability of an email being spam given that it contains the word "Offer" is:

$$P(\text{Spam} \mid \text{Offer}) = \frac{P(\text{Offer} \mid \text{Spam}) \cdot P(\text{Spam})}{P(\text{Offer})}$$

expanding the bottom marginal as the known joint distribution (marginalization in reverse):

$$P(\text{Offer}) = P(\text{Offer} \mid \text{Spam}) \cdot P(\text{Spam}) + P(\text{Offer} \mid \neg\text{Spam}) \cdot P(\neg\text{Spam})$$

Plugging in the given probabilities we get $(0.8 * 0.3)/(0.8 * 0.3 + 0.2 * 0.7) \approx 0.63$

b) Now, to calculate the probability of an email being spam given that it contains both "Offer" and "Urgent", we use **Bayes' Theorem** with the antecedent being the joint distribution of P(Offer,Urgent):

$$P(\text{Spam} \mid \text{Offer}, \text{Urgent}) = \frac{P(\text{Offer}, \text{Urgent} \mid \text{Spam}) \cdot P(\text{Spam})}{P(\text{Offer}, \text{Urgent})}$$

Since "Offer" and "Urgent" are assumed to be independent, we can factorize their joint probabilities as:

$$P(\text{Offer}, \text{Urgent} \mid \text{Spam}) = P(\text{Offer} \mid \text{Spam}) \cdot P(\text{Urgent} \mid \text{Spam})$$

and

$$P(\text{Offer}, \text{Urgent}) = P(\text{Offer} \mid \text{Spam}) \cdot P(\text{Urgent} \mid \text{Spam}) \cdot P(\text{Spam}) + P(\text{Offer} \mid \neg\text{Spam}) \cdot P(\text{Urgent} \mid \neg\text{Spam}) \cdot P(\neg\text{Spam})$$

Substituting these into Bayes' Theorem:

$$P(\text{Spam} \mid \text{Offer}, \text{Urgent}) = \frac{(0.8 \cdot 0.6) \cdot 0.3}{(0.8 \cdot 0.6) \cdot 0.3 + (0.2 \cdot 0.1) \cdot 0.7} \approx 0.91$$

B-2 Maximum Likelihood Estimation (5p)

Suppose you are building a machine learning model to predict the probability of a coin landing heads up. You think that the coin may have a bias, where the probability of heads θ is in the range $0 \leq \theta \leq 1$. You perform an experiment where you flip the coin 10 times, and it lands heads 7 times.

a) **Write** the likelihood function $L(\theta)$ for the observed data above. Hint: The probability of a coin landing heads is θ, which means that the probability of tails is... (2p)

b) **Derive** the maximum likelihood estimate (MLE) for the parameter θ. Show your calculations (2p)

c) What is the relation between the MLE and the true value of the parameter? (1p)

Solution:

a) **Likelihood Function:**

$$L(\theta) = \theta^7(1-\theta)^3$$

This is the likelihood of observing 7 heads and 3 tails, given the probability θ for heads.

b) **Deriving the MLE:**

We maximize the log-likelihood function:

$$\log L(\theta) = 7\log\theta + 3\log(1-\theta)$$

Taking the derivative with respect to θ and setting it to 0:

$$\frac{d}{d\theta}\left(7\log\theta + 3\log(1-\theta)\right) = \frac{7}{\theta} - \frac{3}{1-\theta} = 0$$

Solving for θ:

$$\theta = \frac{7}{10} = 0.7$$

c) MLE only *converges* to the true value as n $\to \infty$ (and standard i.i.d. conditions). For small n it may not be accurate.

B-3 Maximum a Posteriori Estimation (3p)

Consider a simple linear regression model given by:

$$y_i = wx_i + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma^2)$$

where y_i is the observed output, x_i is the input feature, w is the parameter to be estimated, and ϵ_i is Gaussian noise with mean 0 and variance σ^2.

Suppose you have the following prior distribution for the parameter w:

$$p(w) \sim \mathcal{N}(0, \tau^2)$$

where τ^2 is the prior variance. You are given N observations $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$.
Hint: The pdf of a variable x with a Gaussian (Normal) distribution has the form

$$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}. \tag{1}$$

a) **Derive** the **Maximum A Posteriori (MAP)** estimate for the parameter w. (2p)

b) **Compare** the MAP estimator for w to the Maximum Likelihood Estimator (MLE) as $\tau \to \infty$. (1p)

Solution:

a) **Deriving the MAP Estimate:**
Using Bayes' theorem, the posterior distribution is:

$$p(w \mid \mathcal{D}) \propto p(\mathcal{D} \mid w) \cdot p(w)$$

The likelihood is:

$$p(\mathcal{D} \mid w) \propto \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{N} (y_i - wx_i)^2\right)$$

The prior is:

$$p(w) \propto \exp\left(-\frac{w^2}{2\tau^2}\right)$$

Thus, the log-posterior is:

$$\log p(w \mid \mathcal{D}) = -\frac{1}{2\sigma^2} \sum_{i=1}^{N} (y_i - wx_i)^2 - \frac{w^2}{2\tau^2} + C$$

Taking the derivative with respect to w and setting it to 0:

$$\frac{d}{dw}\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{N} (y_i - wx_i)^2 - \frac{w^2}{2\tau^2}\right) = 0$$

Simplifying:

$$w\left(\frac{1}{\sigma^2} \sum_{i=1}^{N} x_i^2 + \frac{1}{\tau^2}\right) = \frac{1}{\sigma^2} \sum_{i=1}^{N} x_i y_i$$

The MAP estimate is:

$$\hat{w}_{\text{MAP}} = \frac{\sum_{i=1}^{N} x_i y_i}{\sum_{i=1}^{N} x_i^2 + \frac{\sigma^2}{\tau^2}}$$

b) Difference Between MAP and MLE:
The MLE for w is:

$$\hat{w}_{\text{MLE}} = \frac{\sum_{i=1}^{N} x_i y_i}{\sum_{i=1}^{N} x_i^2}$$

The MAP estimate includes a regularization term $\frac{\sigma^2}{\tau^2}$, which pulls the estimate towards 0 (the mean of the prior). In short, this means that as $\tau \to \infty$, the MAP estimate will converge to the MLE estimate.

C Graded problems

Potential inquiries to be addressed to Jörg Conradt.

C-1 Multiple-Choice: Support Vector Machine (1p)

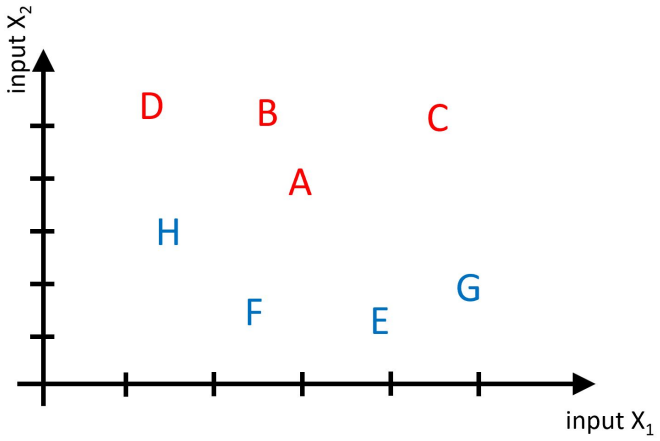
Do not justify your answer. Instead, select exactly **one** option of (1.), (2.), or (3.).
Complete the following sentence: Out of all hyperplanes which solve a classification problem, the one with widest margin will probably ...

1. ... compute fastest.
2. ... have the smallest number of parameters.
3. ... generalize best.

Solution: 3

C-2 Support Vector Classification (3p)

The following diagram shows a small data set consisting of four RED samples (A, B, C, D) and four BLUE samples (E, F, G, H). This data set can be linearly separated.

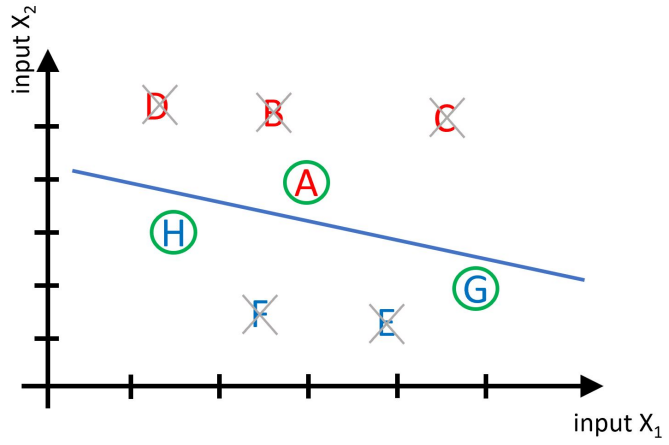


a) We use a linear support vector machine (SVM) without kernel function to correctly separate the BLUE and the RED class. Which of the data points (A-H) can be removed for training without changing the resulting SVM decision boundary? Name the point(s) that can be removed, and justify with **KEYWORDS** (or very short sentences). (2p)

b) Assume someone suggests using a non-linear kernel for the SVM classification of the above data set (A-H). Give one argument AGAINST using non-linear SVM classification for such a data set. **USE KEYWORDS!** (1p)

Solution:

a) The blue line shows the linear decision boundary between the two classes.



We can remove data points: **B, C, D, E, and F**.

Justification: the other points (A, G, H) are closest to the decision boundary. If one of those points gets removed, the decision boundary changes. Therefore, those points are the required support vectors. We can only remove the NON-support vectors without changing the decision boundary.

b) at least one of:

-) More computing resources required (more expensive).

-) slower processing.

-) The algorithm is more difficult to implement.

C-3 Multiple-Choice: Artificial Neural Networks (1p)

Do not justify your answer. Instead, select exactly **one** option of (1.), (2.), or (3.).
Error-Backpropagation-Training for Neuronal Networks requires the following:

1. Gaussian distributed training data and a fast computer.
2. labeled training data and differentiable neuron activation functions.
3. a single input and a single output to the network.

Solution: 2

C-4 Artificial Neural Networks (3p)

Consider the training data in the table, where $+$ means a positive sample and $-$ a negative.

x_1	x_2	Class
4	4	$+$
4	-2	$+$
2	0	$-$
0	2	$-$
-3	-3	$+$
-3	4	$+$

a) What is the minimum number of *layers* needed for an artificial neural network to correctly classify all these points? Motivate your answer **IN KEYWORDS**. (2p)

b) What should the number of input and output nodes be in this network (assume you want a minimal working network)? Motivate your answer **IN KEYWORDS**. (1p)

Solution:

a) Two layers are needed and sufficient. The points are not linearly separable, so one layer is **not** sufficient. Two layers can solve any separation problem.

b) Two input nodes and one output node (given by the data table).